Press release

Tokyo, July 2, 2025

Siemens Energy's Electrolyzer Awarded to Decarbonize Semiconductor Sector in Japan

A notable green hydrogen project in Japan is being equipped with an electrolysis system from Siemens Energy. The project, set to decarbonizing the production of quartz glass for the semiconductor industry in Tamura, Fukushima Prefecture, lays the foundation for the development of a regional hydrogen economy. It also marks the entry of one of the world's leading energy technology companies into the Japanese hydrogen market, highlighting the company's commitment to advancing clean energy solutions.

The project is a joint effort by Tomoe Shokai, Himeji RIKA, the Yamanashi Prefectural Enterprise Bureau, and Yamanashi Hydrogen Company (YHC), and is supported by the New Energy and Industrial Technology Development Organization (NEDO). It aims to establish a localized hydrogen ecosystem powered by renewable electricity generated within Japan. By leveraging green hydrogen in industrial manufacturing, the initiative supports the Ministry of Economy, Trade and Industry (METI)'s roadmap for industrial decarbonization and regional energy resilience.

Siemens Energy's Proton Exchange Membrane (PEM) electrolyzer technology is ideal for integration with renewable energy thanks to its flexible controllability, fast response times, and compact footprint. A PEM electrolyzer uses electricity from renewable sources like solar or wind to split water into hydrogen and oxygen. Its design enables rapid start-up and dynamic operation, making it especially suited for environments like Japan's, where grid-responsive and space-efficient solutions are essential for integrating variable renewable energy into industrial applications.

The system will produce up to 1,900 metric tons of green hydrogen annually – enough to remove over 4,000 gasoline-powered vehicles from Japan's roads. This green hydrogen, along with oxygen, will be used in the manufacturing of quartz glass at Himeji RIKA's facility in Tamura City, enabling a cleaner production process for components essential to Japan's semiconductor industry. Surplus hydrogen will also be supplied to nearby users, laying the foundation for a regional hydrogen supply network.

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Backed by its gigawatt-scale electrolyzer factory in Berlin, the company brings industrial-scale manufacturing expertise and innovation to Japan's growing hydrogen economy.

"Japan has set a bold vision for its energy future, and we are proud to support it," said Russell Cato, Managing Director of Siemens Energy Japan. "Our entry into the Japanese market reflects both our confidence in the country's green hydrogen potential and our commitment to enabling sustainable, high-tech industries like semiconductors through world-class electrolyzer solutions."

The project aligns with Japan's national hydrogen strategy, positioning Tamura City as a key player in developing practical, scalable applications of hydrogen in industrial processes. It also demonstrates how global-local collaboration can drive innovation in one of the world's most technically advanced and sustainability-focused markets.

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Its wind power subsidiary Siemens Gamesa makes Siemens Energy a global market leader for renewable energies. An estimated one-sixth of the electricity generated worldwide is based on technologies from Siemens Energy. Siemens Energy employs around 101,000 people worldwide in more than 90 countries and generated revenue of €34.5 billion in fiscal year 2024. www.siemens-energy.com

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